These requirements are often in conflict with one another. Real-time streams need high-throughput writes and low-latency access, while historical analytics require efficient bulk reads across years of data. Metadata benefits from relational integrity, media files need object storage for large files, and ML models require feature stores for scalable predictions and retraining. No single storage technology handles all requirements effectively. Modern digital twin architectures use *polyglot persistence*, deploying multiple storage systems optimized for specific data types and access patterns. This maximizes performance while controlling costs, though it requires understanding each storage technology's strengths and limitations.

4.5.1 Timeseries databases

Timeseries databases form the backbone of many digital twins, being optimized for the continuous stream of measurements about the changing nature of the physical system that arrive over time. There are many specialized timeseries databases available today, including InfluxDB, Kdb+, and Amazon Timestream, which is available as a managed service within AWS. TimescaleDB is an open source extension to the PostgreSQL relational database optimized for timeseries data. Regardless of the specific product these databases share a set of core features:

- *Columnar storage* - organizes data by measurement type rather than by time, enabling efficient compression and analytical queries across timeseries data.
- *Compression* - timeseries data has a number of characteristics that make it easy to compress, including the fact that timestamps always increase and often at regular intervals, consecutive sensor readings may stay the same for a period of time, or only change by a small amount and sometimes gaps in data collection lead to many null values. Techniques such as *delta encoding* stores only the difference between consecutive values, rather than the values themselves. For example, if temperature readings are `25.1, 25.3, 25.2, 25.4`, storing `25.1, +0.2, -0.1, +0.2` might use less space if the deltas are smaller and can be represented with fewer bits.
- *Aggregation* - automatically creates summary statistics at multiple time granularities, transforming raw data using common statistical and temporal functions (mean, min, max) without expensive real-time calculations. Instead of computing daily averages from millions of individual readings, aggregated data provides pre-calculated summaries that enable instant dashboard updates and historical analysis.
- *Time-based partitioning* - distributes data across storage tiers based on age, keeping recent data in high-performance storage while moving historical data to cheaper options.

4.5.2 Analytical data storage

Analytical query patterns in digital twins typically involve scanning large datasets to identify patterns, correlations, or anomalies across multiple dimensions. These queries differ from operational queries that retrieve specific records by key or recent time ranges.